

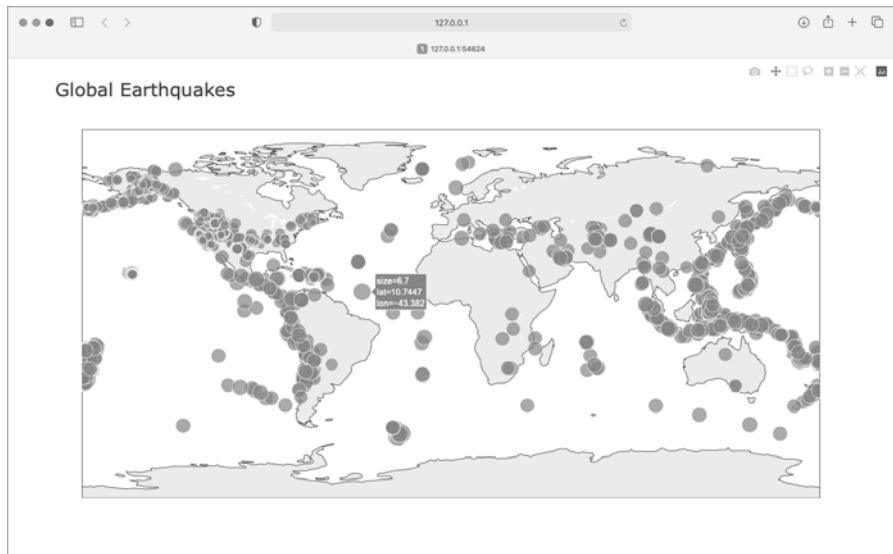


Figure 16-8: The map now shows the magnitude of all earthquakes in the last 30 days.

This map is better, but it's still difficult to pick out which points represent the most significant earthquakes. We can improve this further by using color to represent magnitudes as well.

Customizing Marker Colors

We can use Plotly's color scales to customize each marker's color, according to the severity of the corresponding earthquake. We'll also use a different projection for the base map.

```
eq_world
_map.py  --snip--
fig = px.scatter_geo(lat=lats, lon=lons, size=mags, title=title,
    ❶ color=mags,
    ❷ color_continuous_scale='Viridis',
    ❸ labels={'color': 'Magnitude'},
    ❹ projection='natural earth',
)
fig.show()
```

All the significant changes here occur in the `px.scatter_geo()` function call. The `color` argument tells Plotly what values it should use to determine where each marker falls on the color scale ❶. We use the `mags` list to determine the color for each point, just as we did with the `size` argument.

The `color_continuous_scale` argument tells Plotly which color scale to use ❷. *Viridis* is a color scale that ranges from dark blue to bright yellow, and it works well for this dataset. By default, the color scale on the right of the map is labeled *color*; this is not representative of what the colors actually mean. The `labels` argument, shown in Chapter 15, takes a dictionary as a value ❸. We only need to set one custom label on this chart, making sure the color scale is labeled *Magnitude* instead of *color*.